ÊŒ®ÌÜ® ÉüíË ¿Ê¬Í — Namma Trip

AI-Powered Social Travel Planning Platform for Karnataka

Track: AI Travel Planner for Karnataka

Developer: Drushya M R (drushya-mr | drushyamr2@gmail.com)

Live Website: https://namma-trip-beige.vercel.app

GitHub Repo: https://github.com/drushya-mr/namma-trip

Backend API: https://namma-trip.onrender.com

KEY HIGHLIGHTS:

- 31 / 31 Karnataka Districts Covered (84 Curated Places)
- 100% Kannada-First Experience with 155 Identical i18n Keys
- Strict Grounded AI Itineraries (Zero Hallucination of Fees or Stops)
- Privacy-First Canvas EXIF/GPS Stripper & 9:16 Instagram Story Card Generator
- 21/21 Automated Tests Passing across Integration, Grounding, and Validation

The Problem: Fragmented & Misleading Tourism

Critical barriers faced by travelers across Karnataka

1. Digital Language Barrier: Most travel apps default to English; authentic Kannada terminology and localized context are neglected.
2. AI Travel Hallucinations: Generic LLMs recommend geographically impossible routes (e.g. Bidar to Mangaluru on Day 1).
3. Fabricated Pricing & Timings: Outdated or AI-invented ticket prices and temple timings cause real financial and travel distress.
4. Neglected Micro-Tourism: Tourism focuses solely on Bengaluru/Mysuru/Hampi, leaving 28 other rich districts undiscovered.

Our Solution: Namma Trip (ÊŒ®ÌÜ® ÉüÍË ¿Ê¬Í)

A grounded, bilingual, and inclusive travel ecosystem

- Strictly Grounded AI Planner: 9-Cluster geographic adjacency engine prevents zigzag travel. Stops verified against authentic database IDs.
- Zero-Hallucination Integrity: Never guesses prices or hours. Null fees explicitly marked unconfirmed.
- Inclusive 31-District Coverage: Curated 84 cultural, heritage, and nature destinations spanning all 31 districts.
- Interactive Leaflet Maps: Route pins and paths built on OpenStreetMap with zero paid API keys.
- Kannada-First Experience: Defaults to Kannada (kn) on first visit with instant English toggle; 155-key verified parity.
- Privacy & Social Sharing: On-device canvas EXIF/GPS scrubbing + 9:16 Instagram Story card exporter.

Objectives & Target Audience

Designing for local Kannada speakers, youth, families, and global tourists

CORE OBJECTIVES:

- Cultural Preservation: Modernize Kannada digital travel tools with authentic vocabulary and localized descriptions.
- Equitable Micro-Tourism: Showcase hidden spots in Kalyana-Karnataka, Malnad, and Central Karnataka.
- Trust & Reliability: Establish transparent data grounding so tourists never face hidden charges or closed gates.
- Privacy Protection: Ensure user photos permanently purge device GPS coordinates before sharing to social feeds.

TARGET USER SEGMENTS:

- Local Kannada Families & Seniors: Prefer Kannada navigation, family-friendly pacing, and temple guidelines.
- Youth & Backpackers: Need budget transparency, trekking spots, and downloadable 9:16 Instagram Story cards.
- Heritage Tourists: Benefit from bilingual mode, verified pricing, and interactive Leaflet route maps.

Key Features & Capabilities

A complete, mobile-first social travel application built for reliability

- AI Itinerary Generator: 1-7 day customizable trips by city, budget, travel pace, and interests with cluster adjacency routing.
- All 31 Districts Live Explorer: Real-time 31/31 live coverage counter with bilingual search across names, Kannada scripts, and clusters.
- Interactive Route Maps: Numbered daily pins, route bounds, and null-coordinate safety built with Leaflet.
- Dual-Channel Q&A: Grounded Ask-AI with official citations + Community Q&A with answer voting and accepted-answer verification.
- Transparent Verification Badges: Green badge for 12 verified places; amber badge for 72 unconfirmed places.
- Stories & 9:16 Canvas Cards: Social travel stories feed with client-side EXIF/GPS scrubbing and instant story card export.

Full Karnataka Coverage: All 31 Districts

Organized into 9 geographic clusters to guarantee seamless travel planning

31 / 31 DISTRICTS COVERED — 84 CURATED ATTRACTIONS:

1. Bengaluru Belt (6): Bengaluru Urban & Rural, Ramanagara, Chikkaballapura, Kolar, Tumakuru
2. Old Mysuru & Hoysala Heritage (4): Mysuru, Mandya, Chamarajanagar, Hassan
3. Malnad & Coffee Hills (3): Kodagu, Chikkamagaluru, Shivamogga
4. Coastal Karnataka (3): Dakshina Kannada, Udupi, Uttara Kannada
5. Mumbai-Karnataka / North-West (4): Belagavi, Dharwad, Gadag, Haveri
6. Chalukya Heritage & Deccan Sultanate (2): Bagalkote, Vijayapura
7. Kalyana-Karnataka / North-East (4): Kalaburagi, Bidar, Yadgir, Raichur
8. Hampi & Tungabhadra Region (3): Vijayanagara, Koppal, Ballari
9. Central Karnataka (2): Chitradurga, Davanagere

AI Itinerary Generation & Honest Budgeting

Cluster-constrained AI planning with zero financial illusions

- Cluster-Constrained Planning: Filters candidates strictly to starting cluster and adjacent clusters; prevents cross-state zigzagging.
- Flexible Itinerary Generation: Supports 1-7 days, solo/friends/family groups, relaxed (2 stops/day) vs packed (3 stops/day) styles.
- Single-Day Regeneration: Re-roll any individual day in an itinerary without discarding the rest of the plan.
- Grounded Budget Calculation: Only confirmed entry fees are totaled; null fees are explicitly labeled as unverified.
- Unverified Warning Banner: Any itinerary containing unverified places displays: "É\¨Ëü·ÌÜ È\,Êl¾ÉÌÁ; É\ÆË,µÌ ªÌÜ°Ë\ÇËb ¶Ì ²ÌÜ•É|³Ì ¦Ì<¢Ê¬ŸÌÜŸËü²ÌÜ²".
- Autonomous Local Fallback: Produces authentic itineraries even without external API credits.

Language Accessibility: Built Kannada-First

Prioritizing Kannada language sovereignty in travel technology

- Kannada-Default UX: Opens in Kannada (kn) on first visit for every user, regardless of browser or OS defaults.
- Instant Bilingual Toggle: One-click navbar switch between "É\¨ÌÜ¨Ê " æB $VævÆ—6," v—F, W'6—7FVçB Æö6 Â 7F÷ age.
- 155-Key Symmetrical i18n: Complete linguistic parity enforced by an automated build-time test script (test-i18n-parity.cjs).
- Authentic Vocabulary: Uses natural cultural terms (ÉÌ¿Ë,ÍË,ÆÉ|³Ì Â ªÌÜ°Ë\¾Ë, ¯Ì¼œÊŒÆ, É\°Ì ¨Ëì¡Ì ’ ather than literal machine translation.
- UGC & Review Translation: Built-in translation service converts community posts between Kannada and English on demand.

End-to-End User Experience Workflow

5-step user journey from inspiration to memory sharing

Step 1 — Discover & Explore: Browse 31 districts, filter by cluster or category, check 31/31 live counter.

Step 2 — Configure Preferences: Pick starting gateway city, trip duration (1-7 days), budget, pace, and interests.

Step 3 — Generate & Review: View timeline with Leaflet route map, time slots, entry fees, and single-day regeneration.

Step 4 — Ask AI & Community: Ask questions in Kannada or English; get grounded answers with official government citations.

Step 5 — Share & Export: Upload trip stories with client-side EXIF/GPS scrubbing or download 9:16 Instagram Story canvas cards.

System Architecture & Data Flow

Decoupled, high-performance micro-architecture deployed across Vercel and Render

- Frontend Client (Vercel): React 18 SPA, Vite 6, Tailwind CSS, react-i18next, Leaflet maps, HTML5 Canvas tools.
- Network Proxy Layer: Vercel serverless edge rewrite (/api/* -> Render backend), zero CORS friction.
- Backend API Service (Render): Node.js Express service, Zod schema validation, express-rate-limit, content moderation.
- AI Planning Engine: Google Gemini 1.5 Flash in JSON mode + deterministic heuristic local fallback planner.
- Data Layer: 84 curated places dataset, 9 geographic clusters, PostgreSQL schema (15 tables with RLS), in-memory store.

Technology Stack & Engineering Standards

Production-ready open-source technologies with zero proprietary lock-in

- Frontend: React 18.3.1, Vite 6.4.4, Tailwind CSS 3.4.17, Lucide Icons 1.16, Leaflet 1.9.4, react-i18next 15.4.1.
- Backend: Node.js 24.20.0, Express 4.21.2, Zod 3.24.2, Express Rate Limit 7.5.0, CORS 2.8.5.
- AI & Cloud: Google Generative AI SDK 0.24.0, Supabase JS Client 2.49.1.
- Testing: Node.js built-in test runner (21/21 tests pass), custom i18n parity check script.
- Hosting & CI: Vercel (Edge SPA with proxy rewrites) + Render (Node.js Web Service on port 10000).

Innovation: Grounded AI vs Generic Travel LLMs

Eliminating the risks of hallucinations in public tourism applications

- Destination Integrity: Generic LLMs invent places; Namma Trip enforces candidate IDs strictly from the 84-place database.
- Route Feasibility: Generic LLMs generate random cross-state hops; Namma Trip enforces 9-cluster geographic adjacency.
- Price Transparency: Generic LLMs invent ticket costs; Namma Trip never invents numbers and labels unverified fees.
- Linguistic Authenticity: Generic LLMs output unnatural machine translation; Namma Trip uses human-verified Kannada vocabulary.
- Zero-Cost Reliability: Generic LLMs fail when API credits expire; Namma Trip has an autonomous local fallback planner.

Testing, Deployment & Future Roadmap

21/21 passing automated tests, transparent limitations, and next steps

AUTOMATED TEST RESULTS (100% PASS):

- 21/21 Backend Tests Passing (API Integration, Ask-AI Grounding, Plan Validation, Regression Suite).
- 155/155 i18n Translation Keys with Verified Structural Parity.
- Production Builds: Vite compiles all 2031 modules cleanly with zero errors.

LIMITATIONS & ROADMAP:

- Current Limitation: 12 places verified online; 72 show amber warning badges pending crowdsourced verification.
- Roadmap 1: Transit integration with KSRTC buses and South Western Railway.
- Roadmap 2: Kannada voice search with support for regional dialects (Kundagannada, Arebhashe, North Karnataka).
- Roadmap 3: Certified local guide network with community upvoting.

Conclusion & Project Links

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PROJECT DELIVERABLES & VERIFIED URLS:

- Live Web Application: https://namma-trip-beige.vercel.app
- GitHub Repository: https://github.com/drushya-mr/namma-trip
- Backend REST API: https://namma-trip.onrender.com
- Developer: Drushya M R (drushya-mr | drushyamr2@gmail.com)

HACKATHON ALIGNMENT:

- Complete fulfillment of "AI Travel Planner for Karnataka" problem statement
- 31/31 Karnataka districts represented with 84 curated destinations
- 100% Kannada-first digital accessibility with build-time key parity
- Zero-hallucination grounded AI planning and transparent pricing badges
- Production-deployed, tested, responsive, and ready for public evaluation